A CRM APPLICATION FOR THE CHAI SHOP & CO

6. PROJECT TESTING PHASE

6.1 Purpose of Testing

Testing verifies whether the CRM application operates according to its requirements. The main objectives are:

- Identify errors.
- Verify requirements.
- Validate calculations.
- Check security.
- Test usability.
- Ensure data consistency.

6.2 Testing Types

The project can use:

1. Unit Testing
2. Integration Testing
3. System Testing
4. Functional Testing
5. Usability Testing
6. Security Testing
7. Performance Testing
8. User Acceptance Testing

6.3 Unit Testing

Individual functions are tested separately.

Examples:

- Customer registration function
- Loyalty calculation
- Password validation
- Feedback submission

6.4 Customer Test Cases

Test ID	Test Case	Expected Result
TC01	Add valid customer	Customer created
TC02	Add duplicate customer	Duplicate warning
TC03	Search customer	Correct profile displayed
TC04	Edit customer	Information updated
TC05	Invalid mobile number	Validation message

6.5 Login Test Cases

Test ID	Test	Expected Result
L01	Correct credentials	Login successful
L02	Wrong password	Login rejected
L03	Empty fields	Validation message
L04	Logout	Session ended

6.6 Loyalty Test Cases

Test ID	Test	Expected Result
LO01	Valid purchase	Points added
LO02	Point redemption	Points reduced
LO03	Insufficient points	Redemption rejected
LO04	View balance	Correct balance displayed

6.7 Integration Testing

Integration testing checks whether modules work together.

Example:

Order → Customer → Loyalty

When an order is completed:

1. Customer is identified.
2. Order is stored.
3. Eligible amount is calculated.
4. Loyalty points are calculated.
5. Customer loyalty balance is updated.

Another integration flow is:

Customer → Feedback → Complaint → Resolution

6.8 Security Testing

Security testing verifies:

- Unauthorized users cannot access admin pages.
- Users cannot access functions outside their roles.
- Passwords are handled securely.
- Invalid input cannot corrupt the database.
- Sessions are properly managed.
- Sensitive customer information is not unnecessarily displayed.

6.9 Performance Testing

The system should be tested with increasing numbers of:

- Customers
- Orders
- Feedback records
- Loyalty transactions

The objective is to identify performance problems before deployment.

6.10 UAT

User Acceptance Testing involves representative users performing normal business activities.

Example scenario:

Scenario: Register a new customer.

1. Staff member logs in.
2. Opens Customers.
3. Selects Add Customer.
4. Enters customer information.
5. Saves the record.
6. Searches for the customer.
7. Opens the profile.
8. Confirms information.

6.11 Expected UAT Result

The application should allow staff to complete the process without unnecessary complexity.

6.12 Defect Management

Each defect should be recorded with:

- Defect ID
- Description
- Module
- Severity
- Steps to reproduce
- Expected result
- Actual result
- Status
- Resolution

6.13 Testing Conclusion

After fixing identified defects, regression testing should be performed to ensure that modifications have not introduced new problems.